PROJECT COLLECTION

REZA BABAPIRI

Embedded Systems & Robotics

Sensing, microcontrollers, actuation and functional prototypes

1 TECHNICAL CASE STUDIES

Engineering scope / Method / Results / Visual evidence / Limitations

Technical design, coding and analysis: Reza Babapiri

reza-babapiri.github.io

EMBEDDED HARDWARE · MOBILE ROBOTICS

Autonomous Obstacle-Avoiding Robot

Arduino Uno, ultrasonic sensing, servo scanning and DC motor driving

ENGINEERING QUESTION

Can a compact embedded controller detect obstacles and command a mobile platform using low-cost sensing and actuators?

ROLE & CONTRIBUTION

I built, wired, programmed and tested the physical prototype, which was submitted as a two-person university group assignment. The photograph shows our actual hardware.

ENGINEERING DECISIONS

- Integrated Arduino Uno, HC-SR04 ultrasonic sensing, an actuated scanning servo, L298N motor control and DC geared motors on a tracked platform.
- Implemented ten-reading distance averaging with range filtering, then used a front-distance threshold to activate stop, reverse and turn functions.
- Structured motor actions into reusable embedded functions and monitored sensed distances through serial output.
- Used simple timed motion rather than localization, mapping or route planning, making behavior easier to reproduce and debug.

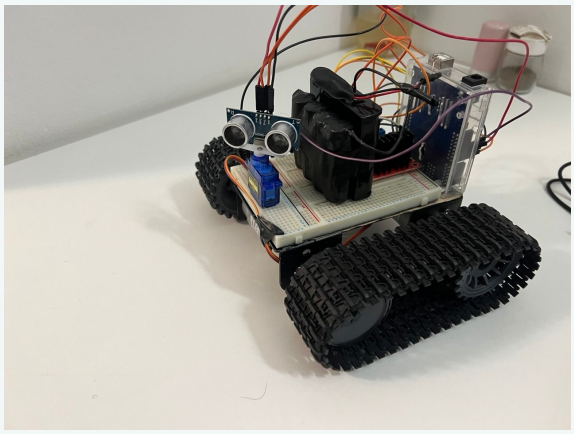
SELECTED ENGINEERING EVIDENCE

The report supplies pin-level wiring and the full Arduino code; the uploaded photograph independently documents the assembled physical prototype. The implemented loop uses a 40 cm threshold.

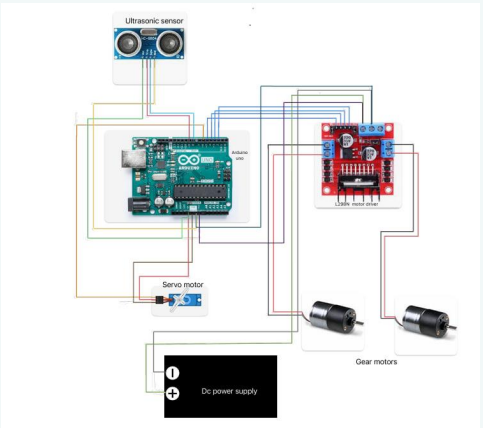
VISUAL EVIDENCE · EMBEDDED HARDWARE · MOBILE ROBOTICS

Autonomous Obstacle-Avoiding Robot

Original images, models and results from the supplied project material



Actual assembled robot photographed by the author: tracked chassis, ultrasonic rangefinder, servo and controller hardware.



Wiring plan from the submitted project report: Arduino Uno, servo, HC-SR04 and motor-driver connections.

RESULTS & ENGINEERING INTERPRETATION

The report supplies pin-level wiring and the full Arduino code; the uploaded photograph independently documents the assembled physical prototype. The implemented loop uses a 40 cm threshold.